

1. Define the term Artificial Intelligence (AI).

Answer:

Artificial Intelligence (AI) is the capability of machines or computers to exhibit human-like intelligence. AI involves technologies that enable machines to perceive, reason, act, and learn like humans. It allows systems to interpret their environment, make decisions, solve problems, and learn from past experiences. AI is already part of our daily lives, powering technologies such as virtual assistants, recommendation algorithms, and facial recognition systems.

2. Classify AI on the basis of capabilities.

Answer:

AI is classified into three categories based on capabilities:

1. **Narrow AI (Weak AI):** AI designed for specific tasks, e.g., Siri, Google Assistant.
 2. **General AI (Strong AI):** AI that can learn and perform any intellectual task like a human (not fully achieved yet).
 3. **Super AI:** Hypothetical AI that surpasses human intelligence and can make decisions beyond human capabilities.
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3. Classify AI on the basis of functionality.

Answer:

AI can be classified into four types based on functionality:

1. **Reactive Machines:** AI that responds to inputs without memory (e.g., IBM Deep Blue).
 2. **Limited Memory AI:** AI that learns from past data to improve decision-making (e.g., self-driving cars).
 3. **Theory of Mind AI:** AI that can understand human emotions and thoughts (future development).
 4. **Self-Aware AI:** AI with human-like consciousness (not yet developed).
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4. Enlist different features of artificial intelligence.

Answer:

AI has the following features:

- Eliminates repetitive tasks
- Uses deep learning and machine learning
- Processes large datasets (Big Data)

- Imitates human cognitive abilities
 - Cloud computing and automation
 - Facial recognition
 - Natural Language Processing (NLP)
 - Computer vision and predictive analytics
 - Robotics and automation
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5. Elaborate Narrow AI (Weak AI).

Answer:

Narrow AI is designed to perform a specific task. It does not have general intelligence and cannot function beyond its programming. Examples include Google Assistant, Alexa, and self-driving cars.

6. Elaborate General AI (Strong AI).

Answer:

General AI aims to perform any intellectual task a human can do. It would possess reasoning, problem-solving, and self-learning capabilities. However, no real-world implementation of General AI exists yet.

7. Elaborate Super AI.

Answer:

Super AI refers to AI systems that surpass human intelligence in all aspects, including decision-making, creativity, and problem-solving. It is currently hypothetical and remains a topic of AI research.

8-9. Elaborate Narrow Reactive Machines and Limited Memory AI.

Answer:

- **Reactive Machines:** AI systems that only respond to inputs without learning from past data (e.g., IBM Deep Blue).
 - **Limited Memory AI:** AI that learns from past experiences and improves decision-making (e.g., self-driving cars).
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10-14. Features of AI (NLP, Computer Vision, Predictive Analysis, Speech Recognition, Personalization).

Answer:

- **NLP:** AI technology that understands and generates human language (e.g., chatbots).
 - **Computer Vision:** AI that interprets and analyzes visual content (e.g., facial recognition).
 - **Predictive Analysis:** AI analyzes past data to predict future trends (used in finance, marketing).
 - **Speech Recognition:** AI converts spoken language into text (e.g., Google Assistant, Siri).
 - **Personalization:** AI customizes experiences based on user behavior (e.g., Netflix recommendations).
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15-17. Importance of AI Features (Predictive Analysis, Decision Making, Automation & Robotics).

Answer:

- **Predictive Analysis:** Helps in decision-making by analyzing data trends (used in business intelligence, healthcare).
 - **Decision Making:** AI assists in making better and faster decisions in various domains like finance, healthcare.
 - **Automation & Robotics:** AI automates repetitive tasks and enhances industrial efficiency (e.g., manufacturing robots).
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18. Evolution of AI (Year-wise).

Answer:

- **1956:** AI officially introduced at Dartmouth Conference.
 - **1970s:** AI designed to mimic human intelligence.
 - **1997:** IBM Deep Blue defeated world chess champion.
 - **2015:** Deep Learning revolutionized AI.
 - **2020:** GPT-3 launched with advanced NLP.
 - **2024+:** AI continues to evolve with advancements in generative models.
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19. Applications of Artificial Intelligence.

Answer:

AI is used in various fields such as:

- Healthcare (disease diagnosis, personalized treatments)
 - Business Intelligence (data analysis, decision-making)
 - Education (personalized learning, automation)
 - Finance (fraud detection, risk analysis)
 - Manufacturing (quality control, automation)
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20. Difference between Machine Learning and Deep Learning.

Answer:

Feature	Machine Learning	Deep Learning
Data Handling	Works on structured data	Works on large unstructured data
Processing	Requires manual feature selection	Self-learns features from data
Examples	Decision trees, SVM	Neural Networks, CNNs

21-22. Supervised and Unsupervised Machine Learning Algorithms.

Answer:

- **Supervised Learning:** Linear Regression, Decision Trees.
 - **Unsupervised Learning:** K-Means Clustering, Principal Component Analysis (PCA).
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23. Reinforcement Machine Learning.

Answer:

Reinforcement Learning (RL) is a type of ML where an agent learns by interacting with the environment using rewards and punishments. Example: AlphaGo.

24. Types of Neural Networks.

Answer:

- Feedforward Neural Networks (FNN)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN).

25-26. Components and Working of AI.

Answer:

Components of AI: Machine Learning, NLP, Computer Vision, Expert Systems.

Working of AI: Uses algorithms and data to make intelligent decisions, often depicted in a flowchart.

27-30. Embedded Systems.

Answer:

- **Definition:** A computer system designed for specific tasks (e.g., microcontrollers).
 - **Applications:** Automobiles, medical devices, industrial automation.
 - **Characteristics:** Real-time performance, low power consumption.
 - **Architecture:** Processor, memory, input/output interfaces.
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31-34. Microcontrollers and ASIC.

Answer:

- **PIC:** A family of microcontrollers used in embedded systems.
- **ARM:** A RISC-based processor with low power consumption.
- **AVR:** A microcontroller designed by Atmel (used in Arduino).
- **ASIC:** A specialized IC designed for specific tasks like AI accelerators.